Evidence-Gated Terrain Admission for a Wheeled Delivery Robot

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Abstract

This document contains the instructions for preparing a valid 2027–2028 Bloomberg Data Science Ph.D. Fellowship Application research proposal. This document conforms to its own specifications and is an example of what your proposal should look like.

The abstract should briefly describe the proposal and make the case for the proposal to non-experts in your particular field. The rest of the proposal should be directed at a technical data science audience.

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1 Introduction

2 Problem and Decision Boundary

A wheeled delivery robot plans over a campus sidewalk graph whose edge labels come from a map prior (OpenStreetMap). Map labels can be stale or wrong; a shortest-path planner that trusts an edge mislabeled `footway` when it actually contains stairs or a curb commits the robot to an untraversable, potentially damaging segment. **Decision:** for each edge e the planner proposes, may the robot commit to traversing e? The policy emits one of four outcomes: **Act** (admit and traverse); **Defer** (unresolved obligation and closer observations remain: continue approach and re-observe); **Repair** (an obligation fails, or remains unresolved with no evidence left: reject e, remove it, replan, and record a proposed map patch); **Handoff** (no admissible route to the goal remains: escalate to a remote operator). Costs are asymmetric: admitting stairs is catastrophic, a false rejection costs a detour. The gate is *fail-closed*: an edge is inadmissible unless evidence affirmatively discharges every obligation for its risk tier. **Research question:** can a calibrated, tiered multi-source gate match a conservative "reject if any source objects" veto on false admission while reducing unnecessary rejection and deferral, including under injected sensor and map faults?

3 Modalities, Data, Typed Claims, Risks, and Obligations

Data. *GND* Liang et al. [2025] (CC0). **Primary:** UMD map 3 (McKeldin Mall), north session: a 24 min, 1.9 km Clearpath Jackal drive (bag chunks 04–14, ∼69 GB) with a 32-beam Ouster LiDAR (10 Hz), wide-angle RGB (640×480, 15 Hz), GPS (5 Hz; reported σ median 9.6 m), wheel/IMU odometry, 1,792 LiDAR-SLAM keyframe poses, and a human-labeled five-class traversability map in which yellow marks *stairs or curbs* Liang et al. [2025] (∼95 candidate stair regions). **Held-out:** Georgetown (GTown; different campus and sensors) as a shift condition. *OSM* OpenStreetMap contributors [2026] (ODbL): footway graph and `highway=steps` tags, cached once as GeoJSON. **Head labels:** weak positives where an OSM stair lies 3–25 m ahead, split by location with buffers; the GND map stays out of training. GPS estimate, ignoring occlusion: ∼20 route sites, ∼840 positive frames. Bags are converted once with the pure-Python `rosbags` library[1]; no ROS or network is needed afterward. **Alignment and ground truth:** the traversability map was annotated on a top-down projection of the LiDAR global map Liang et al. [2025], so map pixels are tied to the SLAM frame by a 2D similarity transform fitted once (ground-removed top view ↔ obstacle layer); GPS is kept out of the ground-truth path and serves only as independent evidence for C2. Observations are associated with edges in this world frame (camera and LiDAR via GND's published extrinsics). An edge is a ground-truth hazard if its footprint intersects yellow: thin yellow lines are labeled curb/step (T1) and yellow regions $\geq A_{\min}$ are labeled stairs (T2). This curb/stairs split is my own definition, not GND's.

Typed claims. Every claim is a typed record with fields `edge`, `predicate`, `value`, `uncertainty`, `source`, `tier`, and `provenance` (sensor, timestamp, frame index, SHA-256 of the cached artifact, and model/checker version). Table 1 gives an example from each source.

Risk tiers. An edge's tier is the *maximum* hazard asserted by any source (escalation only; no source can lower a tier): **T0** flat path; **T1** curb, single step, or curb cut; **T2** stairs. GND notes that wheeled robots cannot traverse stairs or curbs Liang et al. [2025], so the platform step limit h_{lim} is set conservatively below curb height (0.05 m, configurable).

Verification obligations (all must be PASS; any FAIL ⇒ Repair; any UNRESOLVED ⇒ Defer, or Repair if no closer evidence remains): **O1** (all tiers) attribution: C2 confirms the fused GPS+odometry 2σ ellipse intersects only e, which matters because ramps often run parallel to stairs. **O2** (T1+) geometry: C1 reports $h_{\max} < h_{\text{lim}}$ with point coverage $\geq n_{\min}$. **O3** (T2) corroboration: the calibrated $p(\text{stairs}) < \tau_2$ *and* O2 holds on ≥ 2 frames at distinct ranges. **O4** (all tiers) provenance:

[1] `https://gitlab.com/ternaris/rosbags`

Source	Example claim	Role
RGB + frozen embedding	`terrain(e)=stairs, p=0.91`	Calibrated semantic hazard evidence (A1/A2)
LiDAR geometry	`max_step(e)=0.14 m, n_pts=812`	Independent physical evidence (checker C1)
Map prior (OSM)	`tag(e)=footway`	Planning prior; assumed possibly stale
GPS + odometry	`on_edge(e), sigma=2.4 m, d_sep=3.4 m`	Attribution of evidence to one edge (checker C2)

Table 1: Example typed claims by evidence source.

every claim used resolves to a hashed cached artifact. **Risks:** noisy UMD GPS ($\sigma \approx 10$ m; C2 fuses odometry; residual deferral measured); few independent stair sites (counted from SLAM poses; Wilson CIs on T2; achieved calibration bound reported); map-to-SLAM alignment error (residual reported, boundary edges flagged); no camera–LiDAR transforms in the bags (claims fused per edge in the world frame, never per pixel); OSM label errors shared with the map prior (disjoint locations; OSM–GND agreement reported); UMD→GTown shift degrading the camera (caught by O2/O3, and measured).

4 CPU Baseline: Estimated Runtime and Memory

Estimates (4 cores, 16 GB) for ∼3.7k 2 Hz frames (∼2.9k UMD, ∼0.8k GTown); measured values go to `compute.json` at the progress gate. Backbone: frozen DINOv2 ViT-S/14 Oquab et al. [2024] (fallback: MobileNetV3-Small), linear head. Table 2 lists each stage.

Both fit the 4 h / 2 h budgets; cached evaluation artifacts are estimated at <8 GB. **Minimum fallback:** C1 + C2 + map prior, no learned model; still exercises claims, tiers, gate, receipt, invariant, and failure injection end to end.

5 Checkers, Action Gate, Invariant, and Failure Injection

Two independent checkers use disjoint evidence and code paths. **C1, geometric step-height:** fits a local ground plane to LiDAR points on the edge footprint and reports the maximum surface discontinuity; it involves no learning and ignores semantic labels. **C2, localization attribution:** fuses GPS with wheel/IMU odometry in a Kalman filter and uses the fused covariance and graph geometry to decide whether an observation belongs to a single edge. **Gate:** a pure, deterministic function $(\text{claims}, \text{tier}) \mapsto \{\text{Act, Defer, Repair, Handoff}\}$. Missing evidence yields UNRESOLVED, never PASS. Deferral is real in replay: it consumes the next cached observation of e at closer range. **Invariant I1:** *no edge is admitted unless every obligation for its final tier is* PASS*, the tier is at least the maximum asserted by any claim, and every supporting claim's hash resolves.* I1 is checked on every receipt by `validate --strict` and by property-based tests over randomized claim sets. **Repair:** the planner removes the rejected edge and replans; the receipt records a proposed map patch (`tag(e): footway→steps`) with its supporting evidence. **Receipt:** `receipt.json` records config and data-manifest hashes, seed, injected faults, and per-edge claims, obligation states, decision, A3 attribution, and I1 result.

Failure injection (each paired with its expected detection and declared in the receipt): *missing*: camera or LiDAR dropout (⇒ UNRESOLVED, never admit); *corrupted*: blur, low light, glare Hendrycks and Dietterich [2019]; LiDAR thinning; GPS bias of 3–10 m and random-walk drift (⇒ O1 fails); *stale map*: flip k% of curb and stair edges to `footway`; *shifted*: unseen UMD locations, UMD→GTown campus; *adversarial evidence substitution*: flip the map *and* replace stair image crops with flat-sidewalk crops, so that two of three sources agree on "flat" and only C1 stands between the robot and the stairs. **Explicit unwarranted claims:** no claim of real-world deployment safety; of generalization beyond the two evaluated campuses; of GTown stair rates (∼2 regions); of detecting hazards other than steps, stairs, and curbs (ice, potholes, pedestrians); of step heights below LiDAR resolution; or of physical causation. A3 attributions describe the decision procedure, not the world.

6 Metrics, Baselines, Ownership, Timeline, and Fallback

Task performance (reported separately): stairs AUROC/F1 and ECE on held-out GND locations, and C1 agreement with ground-truth step edges. **Policy behavior:** false admission rate (primary; Wilson 95% CI), unnecessary rejection, unnecessary deferral, handoff rate, checker coverage (fraction of admitted edges whose obligations were all discharged by an executable checker), failure detection (fraction of injected faults flagged in the receipt), repair success (replanned route reaches the goal over ground-truth-traversable edges only; overhead relative to an oracle-map route), runtime, and peak RSS. **Baselines:** B0 map-only shortest path; B1 camera-only threshold; B2 conservative veto (reject if any source objects); **B3** the proposed calibrated tiered gate; plus an oracle-map route for reference. **A1/A2/A3:** consume cached embeddings as representation evidence and extend them with a UMD→GTown embedding-shift score that is itself a typed claim (A1); consume calibrated evidence and extend it to multi-source, per-tier thresholds that bound false admission on a calibration split Angelopoulos and Bates [2021] (A2); consume the typed causal claim and extend it to per-decision attribution ("e rejected *because of* C1"), verified exactly by deterministic counterfactual re-runs of the gate with each source ablated (A3). **Track ownership:** solo; I own all tracks.

Timeline. *Oct 31–Nov 6:* lockfile, extraction of UMD north chunks 04–14, map alignment, OSM cache, smoke config. *Nov 7–13:* C1, C2, receipt schema, B0/B2. *Nov 14–20:* embeddings, head, calibration, tiered gate and policy (B1/B3). *Nov 21–30:* injection harness, I1 property tests, measured runtime and memory. **Dec 1 progress gate:** all gate items. *Dec 2–10:* full evaluation, A3 attribution, poster (Dec 11). *Dec 12–15:* report. **Fallbacks:** if map-to-SLAM alignment fails,

Stage	Est. time	Peak RAM
One-time extraction of ∼72 GB of bags (raw images) to a 2 Hz cache	∼65 min	<2 GB
Map alignment, OSM graph, edge association, ground-truth labels	<5 min	<2 GB
Embed ∼3.7k GND frames (∼60 ms/frame)	<5 min	∼3 GB
Head training and temperature calibration Guo et al. [2017]	<3 min	<2 GB
C1 ground fit (RANSAC Fischler and Bolles [1981]) and step height (∼30 ms/frame)	∼5 min	<2 GB
Gate, planner, receipts: 4 policies × 10 conditions × 5 seeds	∼20 min	<3 GB
Full baseline / evaluator path from cache	∼1.7 h / ∼25 min	<6 GB

Table 2: Estimated CPU baseline runtime and memory by stage.

align by hand-picked control points or hand-label ∼200 edges from the GND map; if GND's extrinsics cannot be found, assume coincident base and chassis frames and validate by LiDAR ground-plane height; if the camera head underperforms, report it and rely on the minimum (no-model) fallback; if fused localization still leaves most edges unresolved, relax O1 to parallel-edge ambiguity; if UMD alignment fails, GTown becomes primary.

References

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